

Detection Engineering for Data Loss Prevention: Designing Splunk-Based Alerts and Response Plans for Data Exfiltration Incidents

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User Manual

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1. Purpose of This Manual

This user manual explains how to use the GitHub repository artifacts for the Detection Engineering for Data Loss Prevention project. It is intended for instructors, students, SOC analysts, or technical reviewers who need to inspect, deploy, test, or modify the project files.

GitHub Repository: <https://github.com/rdrake3atangelo/Pedram-and-Robert-Project>

2. Repository Contents

Folder	Description
<code>datasets/synthetic_dataset/</code>	Synthetic dataset containing injected exfiltration scenarios
<code>datasets/benign_dataset/</code>	Benign validation dataset for false-positive testing
<code>detections/</code>	SPL detection rules, entropy macro, and threshold configuration
<code>lookups/</code>	CSV lookup tables used for enrichment and exception handling
<code>playbooks/</code>	Incident response playbooks for UC1 through UC4
<code>evaluation/</code>	Evaluation report, test catalog, and detection tuning log
<code>docs/</code>	Project proposal, design documentation, presentation, and progress reports

3. Prerequisites

A full production deployment would require access to Splunk Enterprise or Splunk Enterprise Security, appropriate permissions to create indexes/lookups/macros/searches, and log sources for proxy/network traffic, DNS queries, email audit events, and DLP policy events. For academic review, the files can be inspected directly from the repository without running Splunk.

Requirement	Details
Splunk indexes	<code>dlp_test</code> , <code>dns_logs</code> , <code>email_logs</code> , <code>dlp_policy</code> or equivalent local index names
Source fields	<code>src_ip</code> , <code>domain</code> , <code>bytes</code> , <code>user</code> , <code>query_name</code> , <code>action</code> , <code>recipient</code> , <code>attachment_size</code>
Lookup files	<code>asset_inventory.csv</code> , <code>authorized_senders.csv</code> , <code>sanctioned_services.csv</code> , <code>unsanctioned_domains.csv</code>
Macros	Threshold macros from <code>thresholds.conf</code> and <code>calc_entropy</code> for UC3

4. Installation and Setup

Use the following process to deploy the project artifacts into a Splunk lab or review environment.

1. Clone or download the GitHub repository.
2. Upload the CSV lookup files from `lookups/` into Splunk Lookup Table Files.
3. Create or update lookup definitions for each CSV file.

4. Create search macros using detections/thresholds.conf. In a Splunk app, thresholds.conf should be placed in the app default directory as macros.conf-compatible content or recreated through Splunk Settings > Advanced Search > Search Macros.
5. Add or recreate the calc_entropy macro from detections/calc_entropy.spl.
6. Load the synthetic and benign datasets into the appropriate Splunk indexes or adapt the SPL searches to local index names.
7. Copy each UC1-UC4 SPL rule into a saved search or ad hoc Splunk search.

5. Detection Rule Usage

Use Case	Detection File	Purpose	Playbook
UC1	detections/uc1_high_volume.spl	Detects high-volume outbound transfer from an internal host to an external domain	PB1
UC2	detections/uc2_unsanctioned.spl	Detects traffic to domains listed in unsanctioned_domains.csv	PB2
UC3	detections/uc3_dns_exfil.spl	Detects DNS tunneling using query count, subdomain length, and entropy	PB3
UC4	detections/uc4_email_exfil.spl	Detects email exfiltration through external sends, recipients, or attachment volume	PB4

Run a detection by opening its SPL file, copying the query into Splunk Search, selecting the desired time window, and executing the search. If local indexes use different names, update the index references before running the search.

6. Expected Alert Fields and Interpretation

Use Case	Important Fields	How to Interpret
UC1	src_ip, hostname, domain, total_bytes, event_count, severity	Investigate large transfer volume and destination legitimacy
UC2	src_ip, hostname, user, domain, flagged_domain, total_bytes	Validate whether the domain is truly unsanctioned or has an approved exception
UC3	src_ip, hostname, query_count, avg_subdomain_len, avg_entropy, sample_queries	Inspect entropy and long random-looking subdomains for tunneling indicators
UC4	user, external_sends, unique_recipients, total_attach_mb, subjects	Assess bulk forwarding, attachment volume, and personal/external recipients

7. Incident Response Playbook Mapping

Each detection has a corresponding playbook. The playbooks guide the analyst through validation, enrichment, containment, escalation, communication, and documentation.

Playbook	Alert Type	Key Response Actions
PB1	High-volume outbound transfer	Block destination, consider host isolation, preserve evidence
PB2	Unsanctioned cloud storage access	Block domain, remove sync tools, revoke tokens, interview user through HR/security

PB3	DNS-based exfiltration	Block apex domain, isolate host, capture memory/disk evidence, escalate immediately
PB4	Email-based exfiltration	Disable forwarding rules, restrict external sending, place mailbox on legal hold

8. Tuning and Maintenance

False-positive control depends on maintaining thresholds and lookup files. The following files should be reviewed regularly.

File	Recommended Review Cadence	Reason
thresholds.conf	Quarterly or after traffic baseline changes	Adjust detection sensitivity without editing core SPL logic
authorized_senders.csv	Quarterly and after marketing/sales process changes	Avoid false positives from approved bulk email senders
sanctioned_services.csv	After SaaS approvals or business process changes	Document approved services and exceptions
unsanctioned_domains.csv	Monthly or after threat intelligence updates	Maintain coverage of risky cloud/file-sharing destinations
asset_inventory.csv	Whenever host ownership or department mapping changes	Keep enrichment accurate for triage

During initial deployment, run searches in monitor-only mode for a baseline period and tune thresholds by user, host, department, and business function. Avoid raising thresholds so high that true exfiltration cases are missed; prefer targeted lookup exceptions when the activity is legitimate and recurring.

9. Testing Procedure

The evaluation/ folder contains the evidence used to validate the framework. The test_catalog.xlsx file defines test cases, eval_results.md summarizes quantitative and qualitative results, and tuning_log.md records threshold and lookup changes.

1. Load or inspect the synthetic dataset and benign validation dataset.
2. Run UC1 through UC4 detections against the expected time windows.
3. Compare actual alerts to the expected results in test_catalog.xlsx.
4. Run searches against benign data to measure false-positive behavior.
5. Apply threshold or lookup changes as needed and record changes in tuning_log.md.
6. Map each true alert to the corresponding playbook and walk through the response steps.

10. Troubleshooting

Issue	Recommended Fix
No results returned	Check index names, source types, and time window. Confirm dataset was ingested.

Lookup errors	Confirm CSV files are uploaded and lookup definitions match the SPL file names.
Macro not found	Create the threshold macros and calc_entropy macro before running detections.
Too many false positives	Review thresholds.conf, authorized_senders.csv, sanctioned_services.csv, and department-specific baselines.
Slow UC3 performance	Consider pre-computing entropy at index time or using summary indexes for high-volume DNS environments.

included.

11. Appendix: File Inventory

Folder	Files
detections/	uc1_high_volume.spl, uc2_unsanctioned.spl, uc3_dns_exfil.spl, uc4_email_exfil.spl, calc_entropy.spl, thresholds.conf
lookups/	asset_inventory.csv, authorized_senders.csv, sanctioned_services.csv, unsanctioned_domains.csv
playbooks/	playbook_pb1.md, playbook_pb2.md, playbook_pb3.md, playbook_pb4.md
evaluation/	eval_results.md, tuning_log.md, test_catalog.xlsx
datasets/	synthetic_dataset/, benign_dataset/